CASE INFORMATION

Case Number:	DF-20260418130703
File Name:	LA_E_1043345.flac
File Type:	Audio
Location:	Uploaded File
Stakeholder:	Investigator
Uploaded:	2026-04-18 13:07:03
Analysis Time:	2026-04-18 13:07:03
Duration:	1.50 sec

PREPROCESSING

Feature:	MFCC (40 coefficients)
Sample Rate:	16,000 Hz
Max Frames:	157
Normalization:	Mean-Std per sample

DETECTION RESULT

VERDICT: FAKE

Real Confidence:	0.00%
Fake Confidence:	100.00%

SPEECH TRANSCRIPTION

Transcribed using OpenAI Whisper (automatic speech recognition)

No speech detected or transcription unavailable.

FORENSIC ANALYSIS & JUSTIFICATION

1. Feature Extraction Rationale

Mel-Frequency Cepstral Coefficients (MFCCs) are used as the primary feature representation because they closely model how the human auditory system perceives sound. MFCCs capture the vocal tract shape and short-term spectral envelope of speech, making them highly discriminative between natural and synthesized audio.

Mean-standard deviation normalization is applied per sample to eliminate recording-level biases caused by variations in microphone gain, room acoustics, or recording equipment. This ensures that the model evaluates spectral shape rather than absolute energy levels.

Fixed-length padding to 157 frames (~2.5 seconds at 16 kHz with a 512-sample hop length) standardizes the input tensor shape required by the CNN architecture. Shorter samples are zero-padded; longer samples are truncated from the right.

2. Audio Behaviour Analysis (Rule-Based)

- Irregular spectral patterns detected across temporal frames, inconsistent with natural human speech production.
- Abrupt or unnatural frequency transitions observed in multiple Mel bands.
- Possible over-smoothing or spectral flattening artifacts -- hallmarks of neural TTS or voice conversion systems.
- MFCC coefficient variance deviates from patterns observed in genuine speech in the training corpus.

3. Model Decision Justification

The model assigns a high Fake confidence of 100.0%, flagging significant anomalies in spectral consistency. These anomalies are characteristic of synthesized or voice-converted audio, where generative models often leave identifiable spectral fingerprints.

4. Confidence Score Interpretation

Confidence scores represent the model's softmax output probabilities, reflecting the degree of certainty in its classification. A score of 100% indicates complete certainty (rarely observed in practice); scores above 70% are generally considered reliable. Scores between 45-55% indicate marginal decisions where both classes exhibit competing feature characteristics.

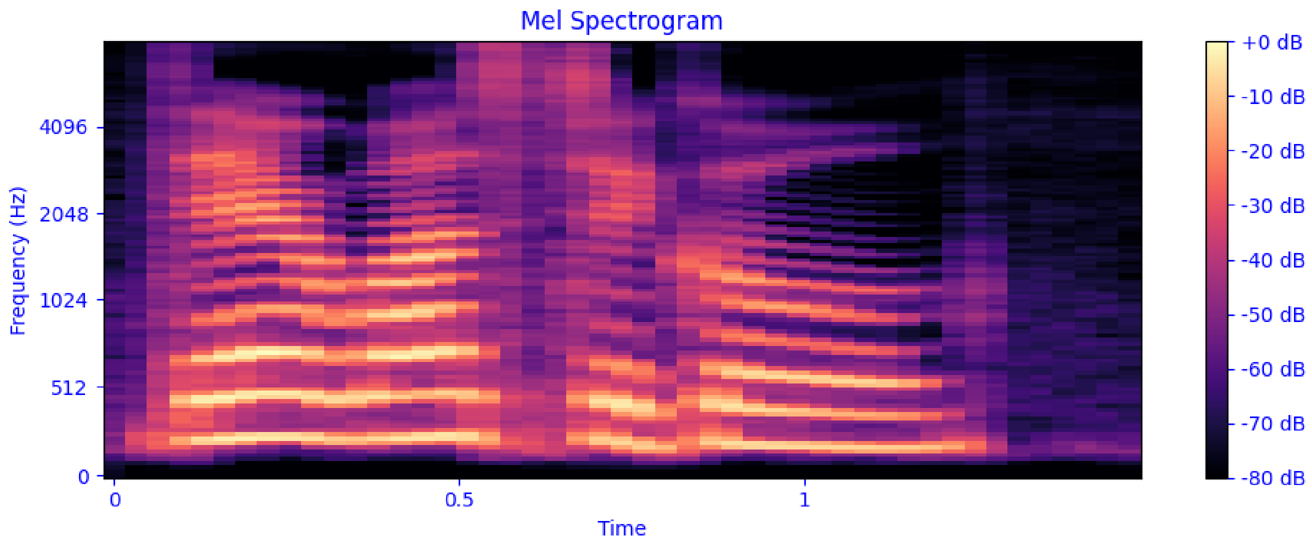
In this case: Real = 0.0%, Fake = 100.0%. These probabilities are derived from patterns learned during training on the ASVspoof 2019 Logical Access (LA) dataset, which includes a wide variety of TTS and voice conversion spoofing attacks.

5. Limitations & Disclaimer

This system is trained exclusively on the ASVspoof 2019 Logical Access dataset and may not generalise to all deepfake types, codecs, or recording conditions. Novel spoofing algorithms not represented in the training data may evade detection. This report is intended as a forensic support tool and must NOT be treated as conclusive legal evidence. Human expert review is strongly recommended before any consequential decision is made based on this output.

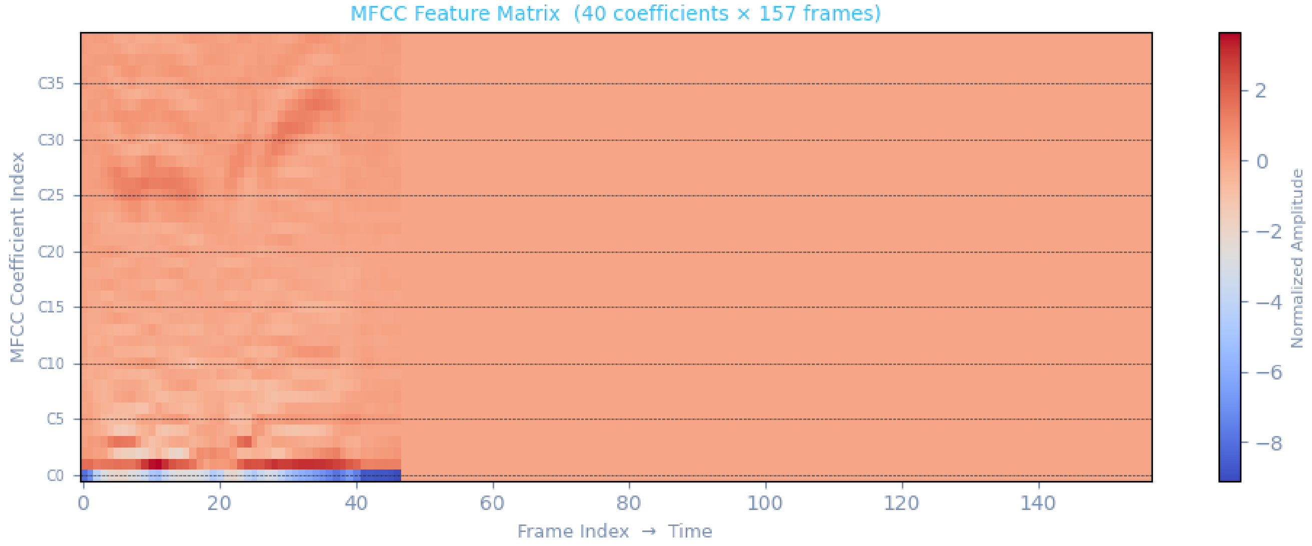
MEL SPECTROGRAM ANALYSIS

Visual representation of frequency energy over time (Mel scale, dB).



MFCC FEATURE ANALYSIS

Heatmap showing 40 MFCC coefficients (rows) across 157 temporal frames (columns). This matrix is the exact input fed to the CNN classifier.



RAW MFCC MATRIX

Shape: 40 coefficients × 157 frames

Overall Statistics

```
Min: -9.083  |  Max: 3.620  |  Mean: -0.000  |  Std: 0.547
```

Per-Coefficient Statistics (Sample: C00-C09)

```
C00: Min=-9.083 Max=0.000 Mean=-1.477 Std=2.577
C01: Min=0.000 Max=3.620 Mean=0.588 Std=1.024
C02: Min=-2.074 Max=1.163 Mean=-0.064 Std=0.502
C03: Min=-0.948 Max=1.921 Mean=0.053 Std=0.341
C04: Min=-1.521 Max=0.470 Mean=-0.129 Std=0.346
C05: Min=-0.840 Max=0.677 Mean=0.014 Std=0.232
C06: Min=-1.041 Max=0.209 Mean=-0.083 Std=0.232
C07: Min=-0.931 Max=0.252 Mean=-0.089 Std=0.226
C08: Min=-0.835 Max=0.267 Mean=-0.072 Std=0.195
C09: Min=-0.552 Max=0.131 Mean=-0.054 Std=0.128
```

(Full matrix available in application interface)